

HAT-P-36b

R-0917

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230516 · created 2026-07-29T08:02:28+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460080.8107 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.046
Transit depth (Rp/Rs)^2	0.00017 +/- 0.012 [%]
Orbital Inclination (inc)	83.18 +/- 2.87
Airmass coefficient 1 (a1)	0.63 +/- 0.34
Airmass coefficient 2 (a2)	0.52 +/- 0.2
Scatter in the residuals of the lightcurve fit is	68.0 %
Best Comparison Star	#2 - [401, 261]
Optimal Aperture	4.77
Optimal Annulus	9.89
Transit Duration (day)	0.074 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.046 vs 0.126 (deviation 1.7σ)
Duration fit vs published	0.074 d vs 0.0925 d (deviation 0.77σ)
Mid-time O-C	-7.3 min
Depth SNR	0.0
Residual scatter	68.0 %

Light curve

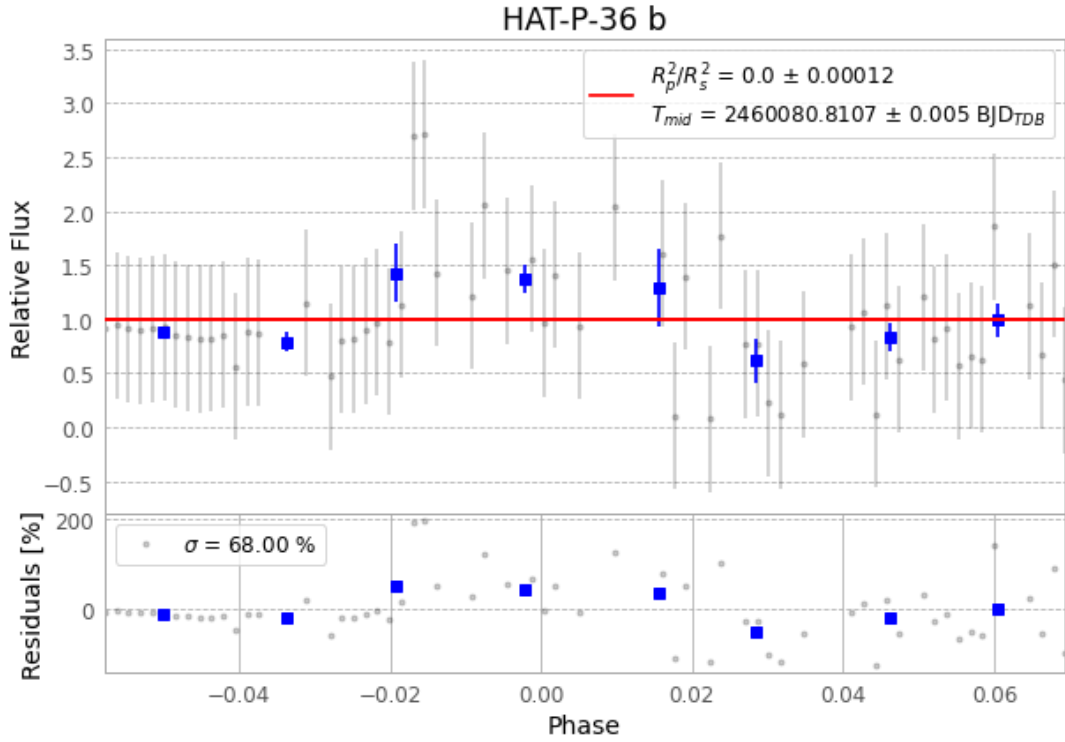


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-05-15.png (SHA-256 ee9cda870fa852ae...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 392], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fd5b0d75c4c0a611...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

82 FITS frames (54 MB), HATP-36230516053317.FITS → HATP-36230516093615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230516.log (3b701e131d89...), FinalLightCurve_HAT-P-36 b_2023-05-15.png (ee9cda870fa8...), FinalLightCurve_HAT-P-36 b_2023-05-15.csv (c3dce0e4f3e8...)

Compute resources

Wall time 205.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 08:02Z.